



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CS&IT
Course Outcome	CO-1

Case Study Topic: Content Metadata Indexing using AVL Tree in MediaVault

Work Done Summary:

MediaVault stores thousands of digital contents such as movies, web series, podcasts, and videos. Efficient retrieval of content metadata is essential for providing fast search functionality to users.

An AVL Tree was implemented to store Content IDs and maintain a balanced indexing structure. This enables efficient searching and retrieval of content records even when the database grows significantly.

Problem Statement

MediaVault continuously receives new content uploads. If content records are stored using a normal BST, sequential insertions may create a skewed tree structure and increase search time.

The challenge is to:

- Maintain balanced indexing.
- Improve content retrieval speed.
- Support efficient insertion and searching.
- Handle large content repositories.

Proposed Solution

Use an AVL Tree to store Content IDs.

Implementation Details:

Algorithm Used

AVL Tree

Operations Performed

1. Insert Content IDs.
2. Perform AVL rotations.
3. Search content records.
4. Display final AVL Tree.
5. Perform inorder traversal.

```

public class ContentAVL {
    public static void main(String[] args) {
        tree.printTree(tree.root,0);

        System.out.println(
            "\nInorder Traversal:");

        tree.inorder(tree.root);

        System.out.println(
            "\n\nSearch Content ID 205 : "
            + (tree.search(tree.root,205)
            ? "Content Found"
            : "Content Not Found"));

        System.out.println(
            "Search Content ID 250 : "
            + (tree.search(tree.root,250)
            ? "Content Found"
            : "Content Not Found"));

        System.out.println(
            "\n=====");

        System.out.println(
            "Content Indexing Completed");

        System.out.println(
            "=====");
    }
}

```

Result: Screenshots/Output

```

=====
FINAL AVL TREE
=====
                208
              207
            206
          205
        204
      203
    202
  201

Inorder Traversal:
201 202 203 204 205 206 207 208

Search Content ID 205 : Content Found
Search Content ID 250 : Content Not Found

=====
Content Indexing Completed
=====

```

- AVL Tree maintained balance successfully.
- Content retrieval was efficient.
- Rotations prevented tree skewness

GitHub Link:

<https://github.com/snehiths399-byte/MediaVault--Digital-content-streaming-and-Analytics-system>

Student Signature	Faculty Signature